

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview	
Objective	The primary objective is to develop an AI-based system that predicts an individual's overall well-being using Machine Learning, enabling users to better understand their health and receive personalized recommendations.
Scope	The project includes data collection, preprocessing, feature selection, machine learning model development, prediction, visualization of results, and deployment through a Flask-based web application for real-time user interaction.
Problem Statement	
Description	Many people find it difficult to evaluate their overall well-being because existing solutions focus on only one aspect of health, such as fitness or mental health. Manual assessments are time-consuming and often lack personalized insights.
Impact	Solving this problem will enable users to monitor their well-being more accurately, improve lifestyle decisions, promote preventive healthcare, and support healthcare professionals with faster assessments.
Proposed Solution	
Approach	Develop a Machine Learning model using Linear Regression to analyze user health and lifestyle information, predict a comprehensive well-being score, and display the results through a Flask web application with graphical visualizations and recommendations.
Key Features	• User-friendly web interface using Flask. • Secure user data collection and validation. • Data preprocessing and feature selection. • Well-being prediction using Linear Regression. • Interactive dashboard with graphs and charts. • Personalized recommendations for improving well-being. • Model saving and real-time prediction using Joblib.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU for model training and prediction	Intel Core i5/i7 or equivalent
Memory	RAM for processing datasets	Minimum 8 GB RAM
Storage	Space for datasets, trained model, and project files	256 GB SSD or higher
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code / Jupyter Notebook
Data		
Data	Source, size, format	Well-Being Dataset (CSV format), containing health, lifestyle, and demographic features used for training and testing the Machine Learning model.